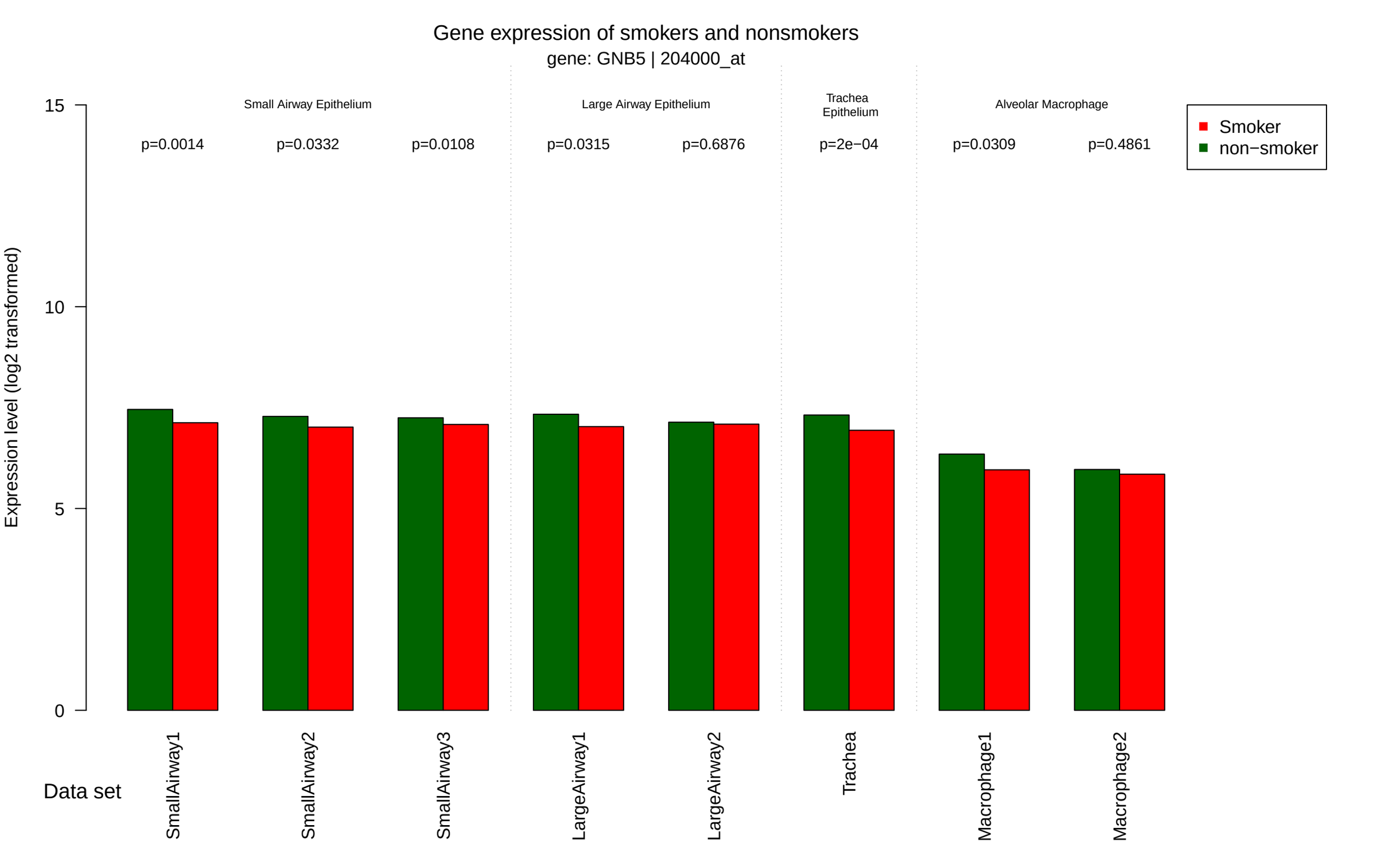
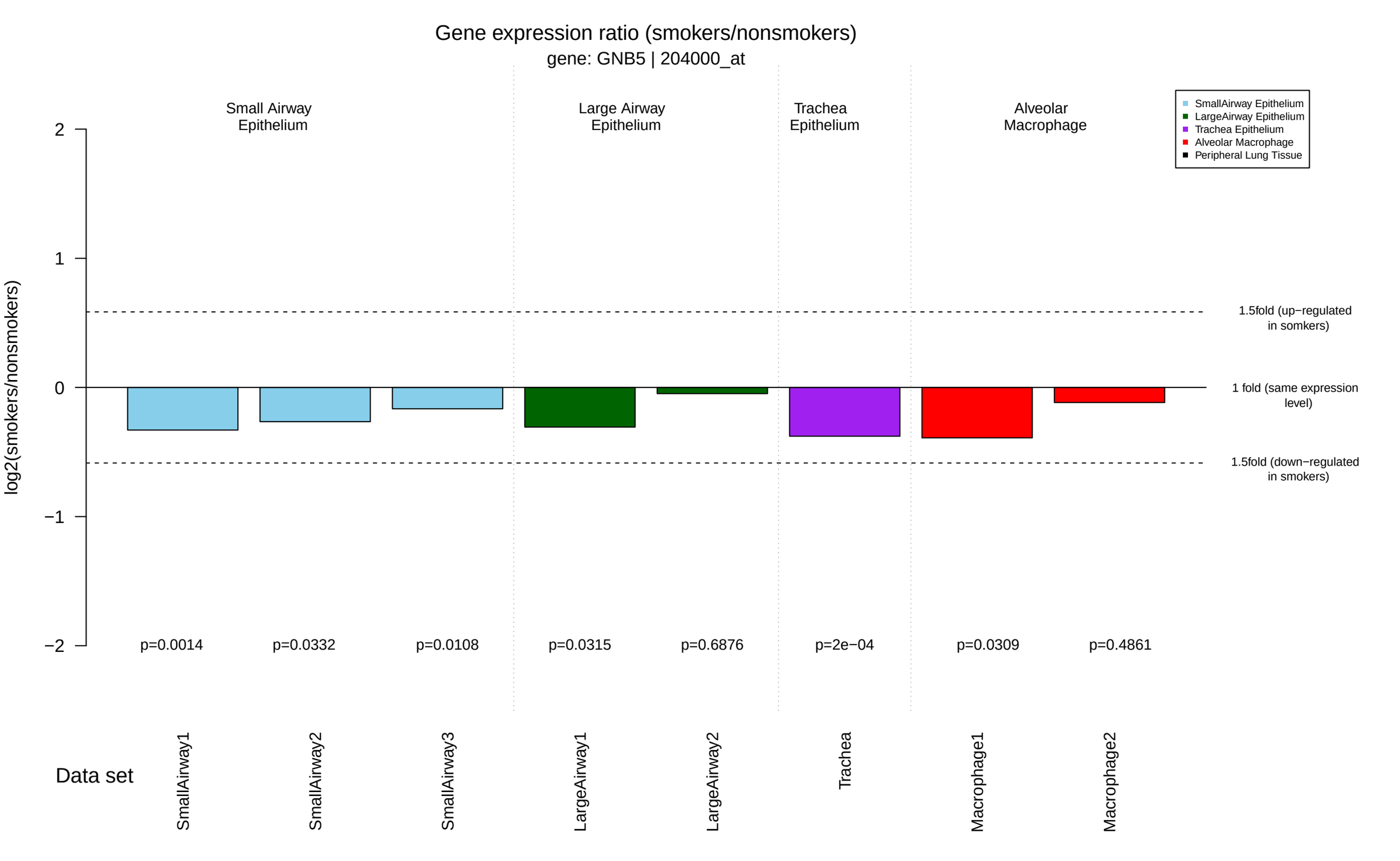
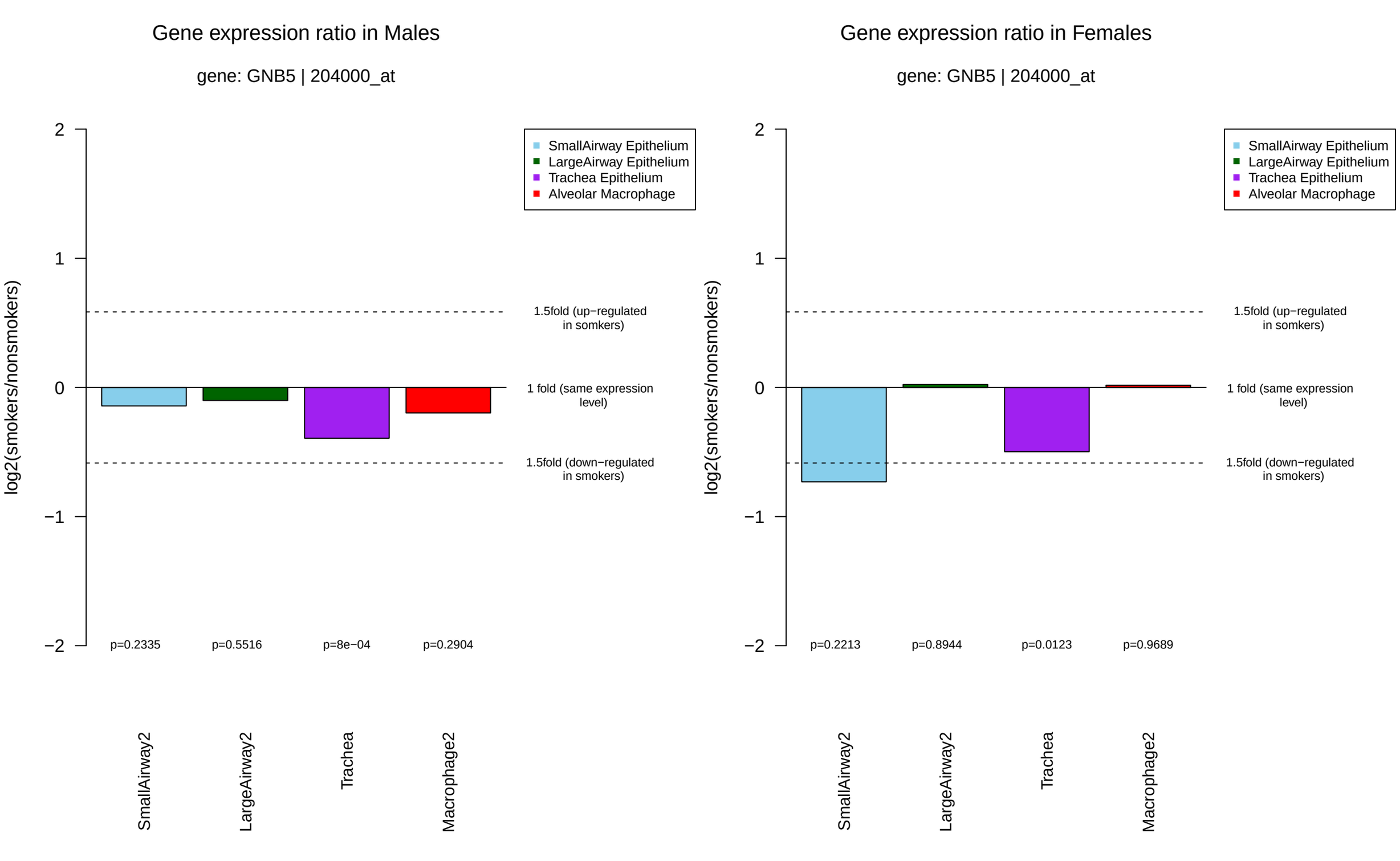
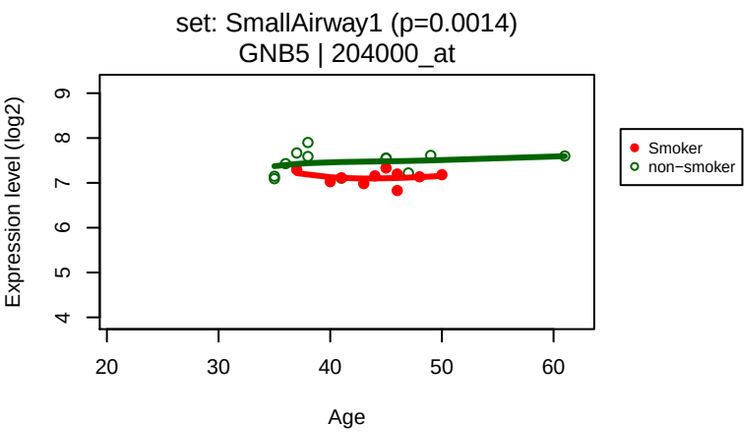
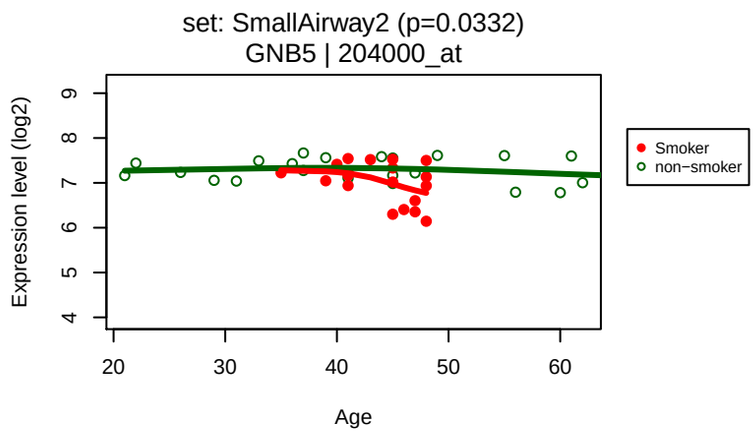
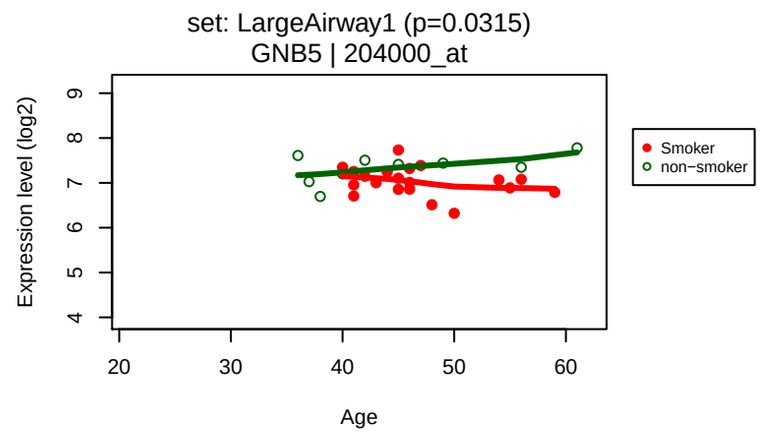

Gene expression of smokers and nonsmokers
gene: GNB5 | 204000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0014
p=0.0332
p=0.0108
p=0.0315
p=0.6876
p=2e−04
p=0.0309
p=0.4861
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

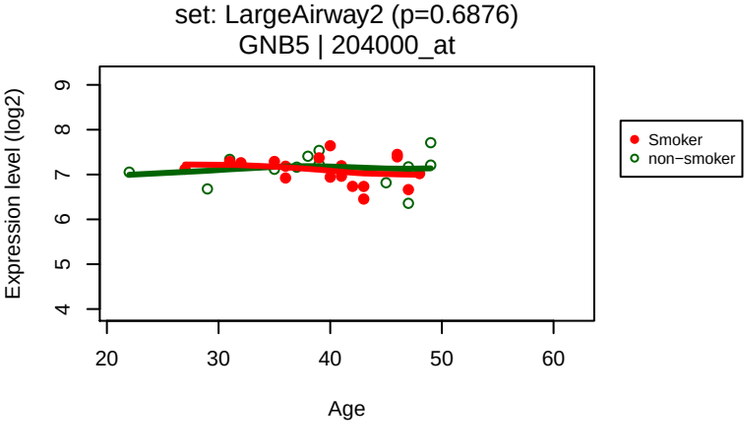

Gene expression ratio (smokers/nonsmokers)
gene: GNB5 | 204000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0014
p=0.0332
p=0.0108
p=0.0315
p=0.6876
p=2e−04
p=0.0309
p=0.4861
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

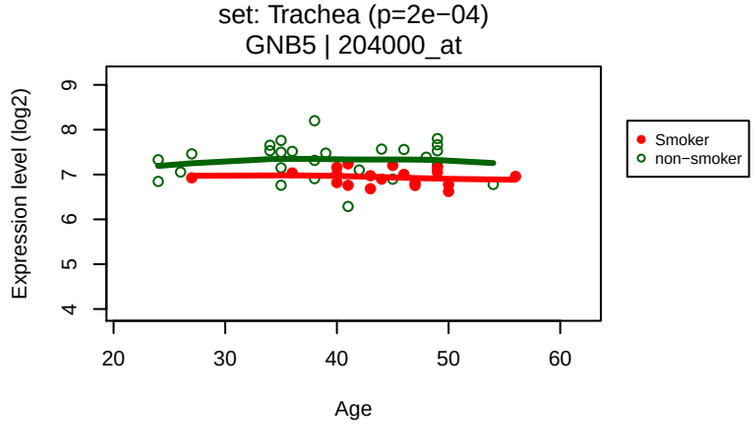

Gene expression ratio in Males
gene: GNB5 | 204000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2335
p=0.5516
p=8e−04
p=0.2904
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNB5 | 204000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2213
p=0.8944
p=0.0123
p=0.9689
SmallAirway2
LargeAirway2
Trachea
Macrophage2

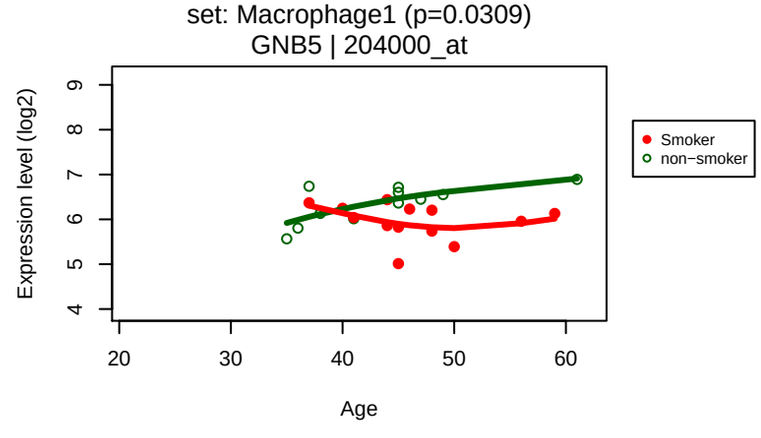

set: SmallAirway1 (p=0.0014)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker

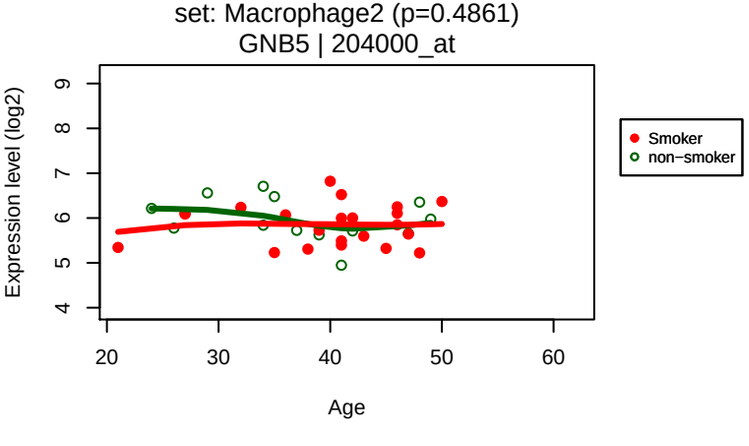

set: SmallAirway2 (p=0.0332)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker

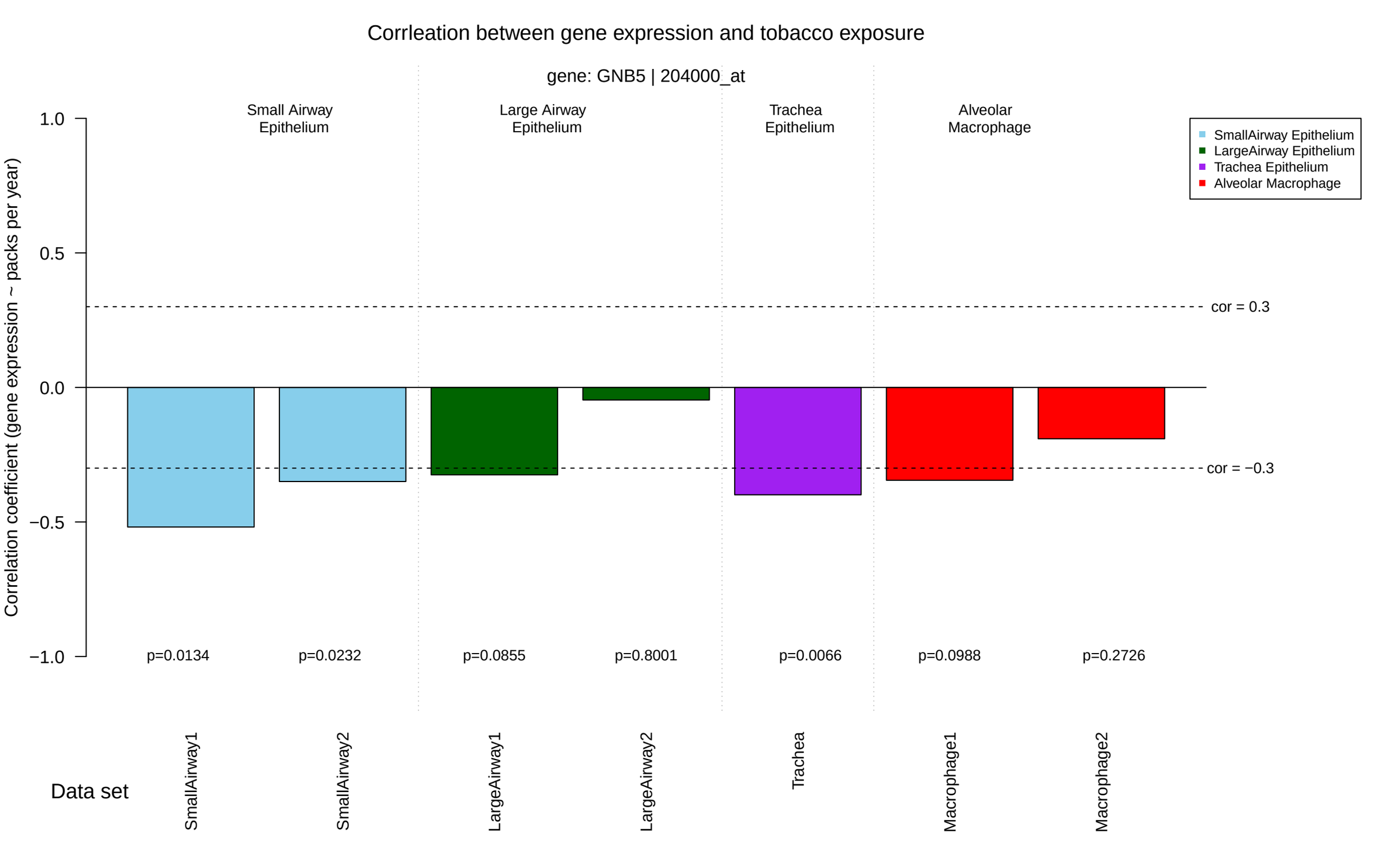

set: LargeAirway1 (p=0.0315)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6876)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0309)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4861)
GNB5 | 204000_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNB5 | 204000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0134
p=0.0232
p=0.0855
p=0.8001
p=0.0066
p=0.0988
p=0.2726
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2